

torch.svd_lowrank#

https://pytorch.org/docs/stable/generated/torch.svd_lowrank.html#torch.svd_lowrank

Description:

Return the singular value decomposition (U, S, V) of a matrix, batches of matrices, or a sparse matrix AAA such that $A \approx U \text{diag}(S) V^H$. In case MM^T is given, then SVD is computed for the matrix $A - MA - MA - M$. The implementation is based on the Algorithm 5.1 from Halko et al., 2009. For an adequate approximation of a k -rank matrix AAA , where k is not known in advance but could be estimated, the number of columns, q , can be chosen according to the following criteria: in general, $k \leq q \leq \min(2k, m, n)$. For large low-rank matrices, take $q = k + 5..10$. If k is relatively small compared to $\min(m, n)$, choosing $q = k + 0.2 \min(m, n)$ may be sufficient.

Function Signature

```
torch.svd_lowrank(A, q=6, niter=2, M=None)[source]#
```

Args::

niter (int, optional): the number of subspace iterations to

M (Tensor, optional): the input tensor's mean of size

References::

Returns:

tuple[torch.Tensor, torch.Tensor, torch.Tensor]

Notes & Warnings

NOTE

Note The implementation is based on the Algorithm 5.1 from Halko et al., 2009.

NOTE

Note For an adequate approximation of a k -rank matrix AAA , where k is not known in advance but could be estimated, the number of QQQ columns, q , can be chosen according to the following criteria: in general, $k \leq q \leq \min(2 * k, m, n)$. For large low-rank matrices, take $q = k + 5..10$. If k is relatively small compared to $\min(m, n)$, choosing $q = k + 0..2$ may be sufficient.

NOTE

Note This is a randomized method. To obtain repeatable results, set the seed for the pseudorandom number generator

NOTE

Note In general, use the full-rank SVD implementation `torch.linalg.svd()` for dense matrices due to its 10x higher performance characteristics. The low-rank SVD will be useful for huge sparse matrices that `torch.linalg.svd()` cannot handle.

Detailed Documentation

Documentation

Return the singular value decomposition (U, S, V) of a matrix, batches of matrices, or a sparse matrix AAA such that $A \approx U \text{diag}(S) V^H$. In case MMM is given, then SVD is computed for the matrix $A - MA - MA - M$.

The implementation is based on the Algorithm 5.1 from Halko et al., 2009.

For an adequate approximation of a k -rank matrix AAA , where k is not known in advance but could be estimated, the number of QQQ columns, q , can be chosen according to the following criteria: in general, $k \leq q \leq \min(2k, m, n)$. For large low-rank matrices, take $q = k + 5..10$. If k is relatively small compared to $\min(m, n)$, choosing $q = k + 0.2 \min(m, n)$ may be sufficient.

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In general, use the full-rank SVD implementation `torch.linalg.svd()` for dense matrices due to its 10x higher performance characteristics. The low-rank SVD will be useful for huge sparse matrices that `torch.linalg.svd()` cannot handle.

A (Tensor): the input tensor of size $(*, m, n)$

q (int, optional): a slightly overestimated rank of A .

`conduct`; `niter` must be a nonnegative integer, and defaults to 2

$(*, m, n)$, which will be broadcasted to the size of A in this function.

Nathan Halko, Per-Gunnar Martinsson, and Joel Tropp, Finding structure with randomness: probabilistic algorithms for constructing approximate matrix decompositions, arXiv:0909.4061 [math.NA; math.PR], 2009 (available at arXiv).

tuple[torch.Tensor, torch.Tensor, torch.Tensor]

